



Semester I Examinations 2017/18

Exam Code(s)	1SD1, 1SD3, 1MF1
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Pier Luca Lanzi
Internal Examiner(s)	*Dr. Michael Schukat

Instructions: Answer two questions from Section A and one question from Section B.

Duration	2 hrs
No. of Pages	6
Discipline(s)	Information Technology
Requirements	None

Release to Library: Yes ☒ No ☐

Section A

Question 1

- a) Distinguish between the *Harvard architecture* and the *von Neumann architecture*. Explain which of the two architectures allow data to be interpreted as program code and visa versa.
3 MARKS
- b) Using an example outline in detail how the *paged memory allocation* scheme works. In your answer refer to the terms, pages, frames, page tables, and demand paging.
8 MARKS
- c) Summarise the 7-state process model and, using an example, show how a process moves through all different states.
6 MARKS
- d) Express the decimal values between 0 and 20 in hexadecimal and binary.
3 MARKS

Question 2

- a) Distinguish between the following addressing modes:
 - Immediate Addressing
 - Direct Addressing
 - Indirect Addressing3 MARKS
- b) Distinguish between *interrupt-driven* and *DMA-driven* data input/output. Why would you choose one over the other?
5 MARKS
- c) What is the purpose of the program counter (PC)?
Outline a situation where, during the execution of a program, the content of the PC is not simply incremented (in order to fetch the next instruction), but replaced with an entirely different address.
4 MARKS
- d) Using code snippets and / or diagrams show how two concurrent processes can use a semaphore to control their access to a non-sharable or critical resource.
8 MARKS

Question 3

- a) Explain the two diagrams below and provide a detailed step-by-step example on the operation of a cache.

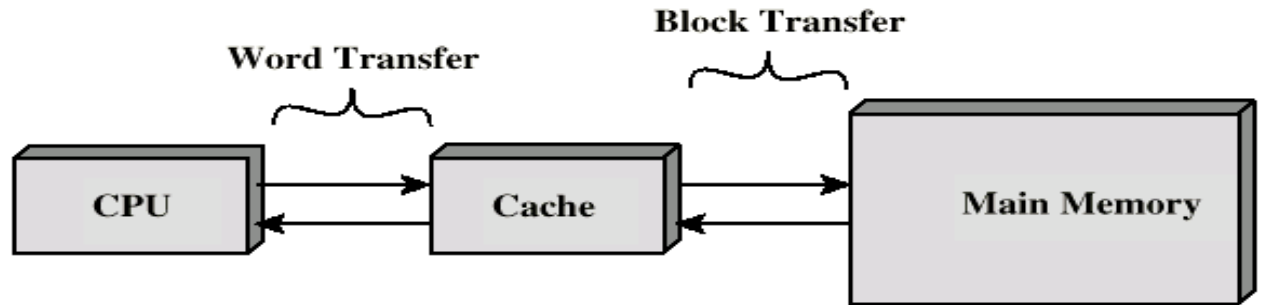


Figure 1: CPU / Cache / Main Memory

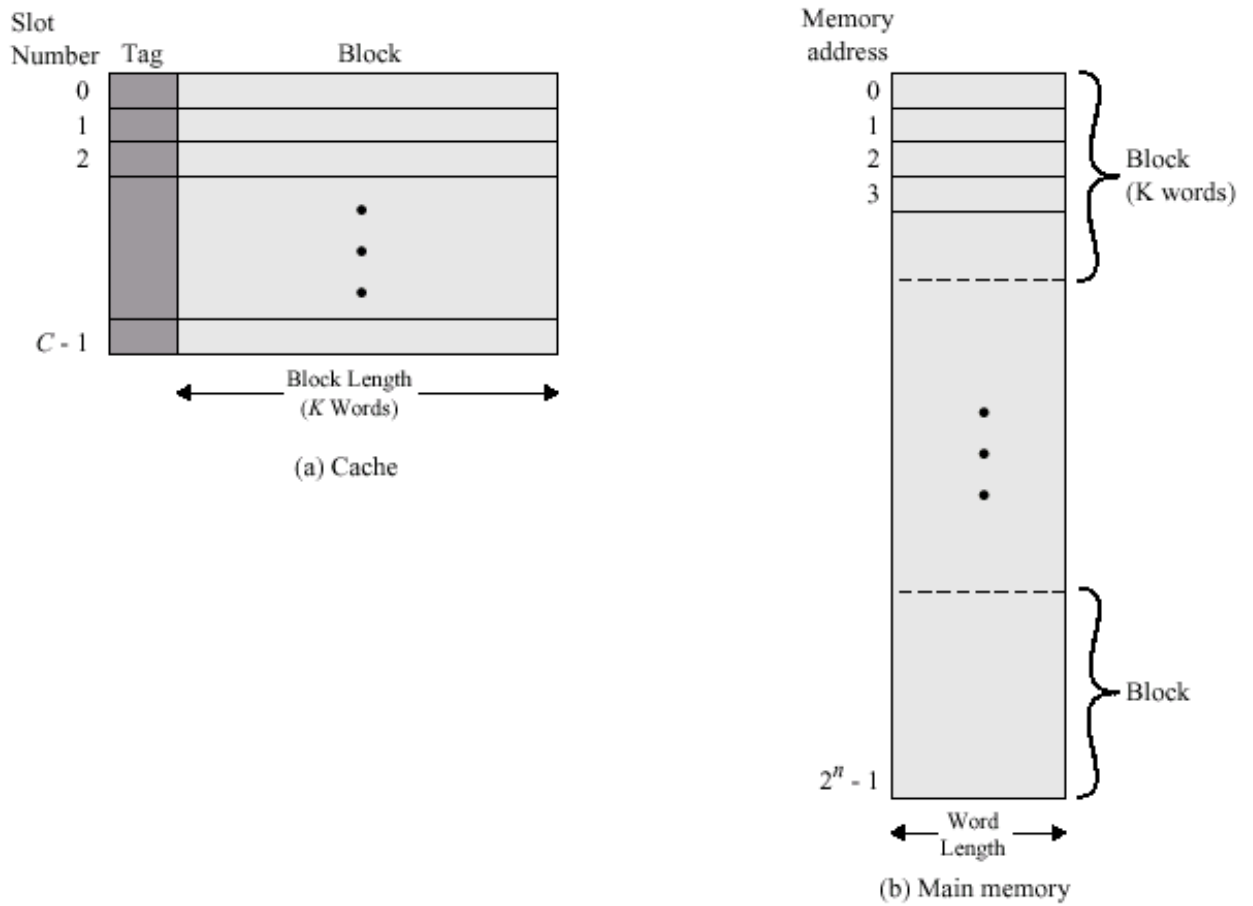


Figure 2: Cache organisation

8 MARKS

- b) Explain the inner workings of a hard disk with multiple platters. In your answer refer to the terms *sector*, *track* and *cylinder*.

4 MARKS

PTO

- c) Using examples distinguish between the three process scheduling algorithms
- first come – first served
 - shortest remaining time
 - round robin.

6 MARKS

- d) Explain why cache memory is based on static rather than dynamic RAM.

2 MARKS

PTO

Section B

Question 4

- a) Summarise and compare the functionality of the layers of the OSI and TCP/IP reference model.

5 MARKS

- b) Discuss the purpose and inner workings of the following protocols:
- Dynamic Host Configuration Protocol (DHCP)
 - Address Resolution Protocol (ARP)

5 MARKS

- c) Using diagrams explain the purpose and inner workings of TCP window management.

8 MARKS

- d) Briefly outline the similarities and differences between a *hub* and a *switch*.

2 MARKS

Question 5

- a) Using the diagrams below explain how the *distance vector routing* algorithm works.
Further on, discuss the *count-to-infinity* problem.

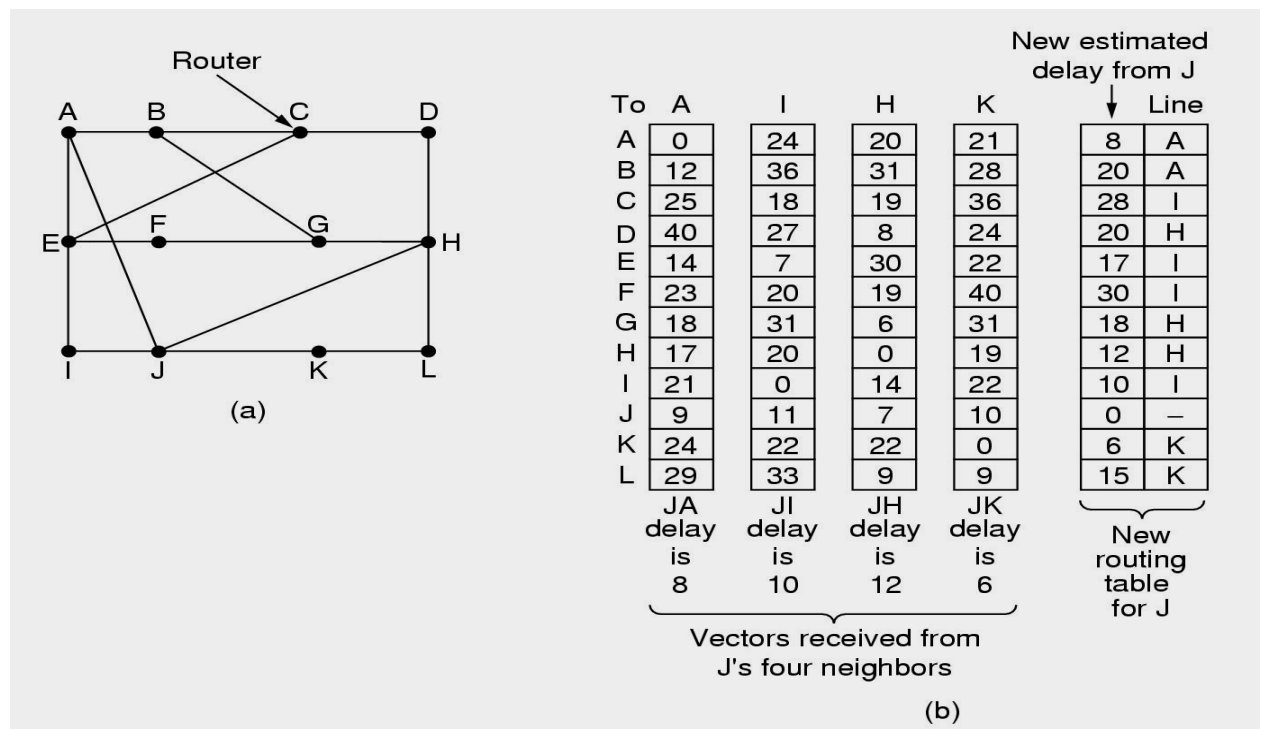


Figure 3: Distance vector routing

5 MARKS

PTO

- b) Using an example outline how client-side packet buffering reduces jitter and hence improves the quality of service.
3 MARKS
- c) Discuss the similarities and differences between the multiple access protocols *pure ALOHA* and *Carrier Sense Multiple Access (CSMA)*. Explain which of the 2 protocols performs generally better.
5 MARKS
- d) Explain how TCP uses a 3-way handshake protocol to establish a connection.
7 MARKS